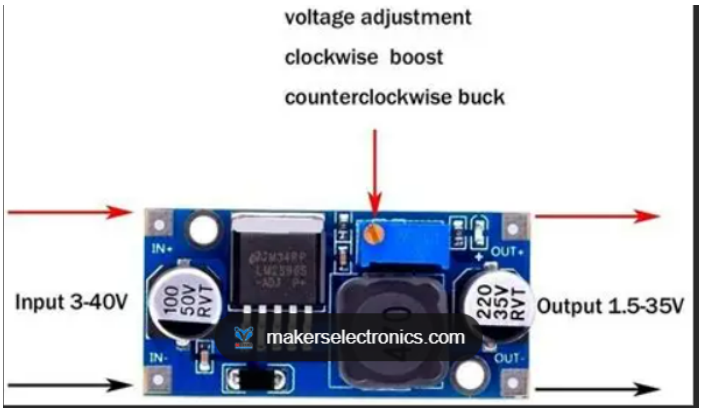
EsiBot Wiring

Component 1 — (3× 18650, 3.7V, 2000mAh) 3S

Pin	Connects to	Component	Notes
(+) positive	VIN	LM2596 DC-DC converter	Raw ~11.1v input to converter
(+) positive	VSS	L298N	Raw ~11.1V direct to motor supply — do NOT regulate
(−) negative	Common GND rail	—	Star point for all grounds

Component 2 — LM2596 DC-DC step-up converter



Pin	Connects to	Component	Notes
VIN	(+) positive	3s battery	Receives raw ~11.1V
GND (input)	Common GND rail	—	—
VOUT	VCC (5V input)	ESP32-CAM	Regulated 5V — set with multimeter before connecting
VOUT	VCC logic supply	L298N motor driver	Logic supply only, separate from VSS
VOUT	VCC (red wire)	MG996R servo	MG996R draws up to 2.5A stall — factor into total current budget
VOUT	VCC	HC-SR04 sensor	Sensor runs on 5V
GND (output)	Common GND rail	—	Input and output GND both go to common rail

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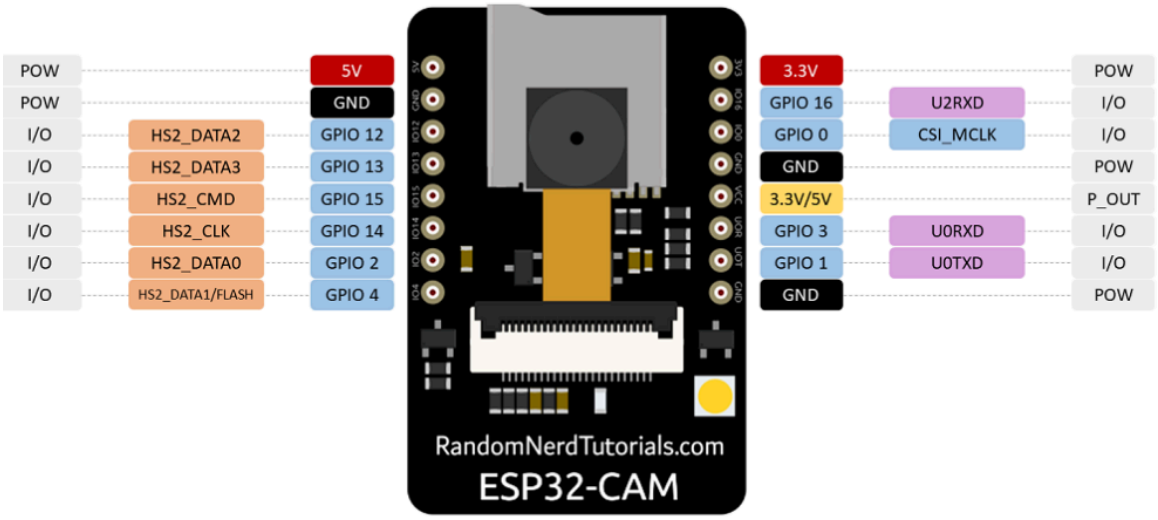
Component 3 — ESP32-CAM (AI-Thinker)



Pin	Connects to	Component	Notes
VCC (5V)	VOUT	LM2596	5V regulated supply
GND	Common GND rail	—	Must share common ground with RPi for UART to work
GPIO1 (U0TXD)	GPIO15 / RXD (Pin 10)	Raspberry Pi 4	UART TX — ESP32 sends camera data to RPi
GPIO3 (U0RXD)	GPIO14 / TXD (Pin 8)	Raspberry Pi 4	UART RX — ESP32 receives commands from RPi

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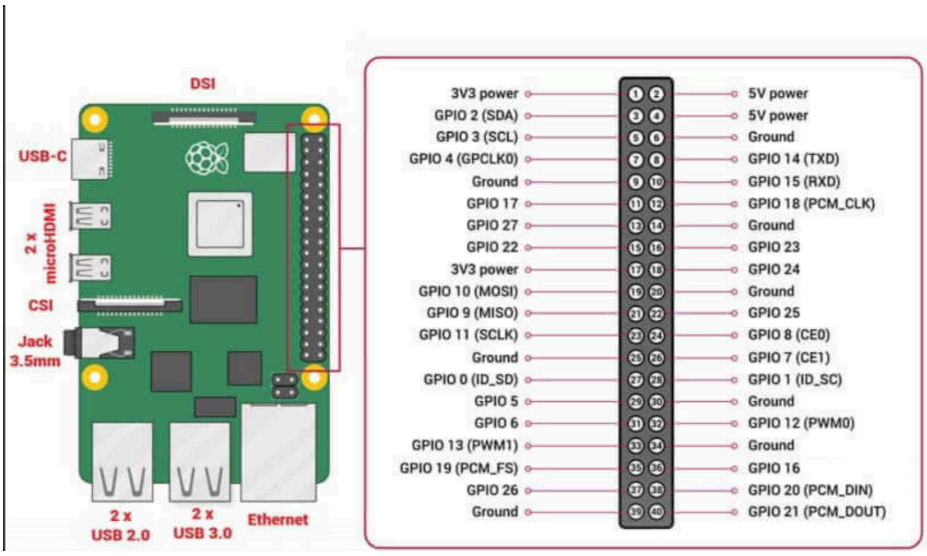
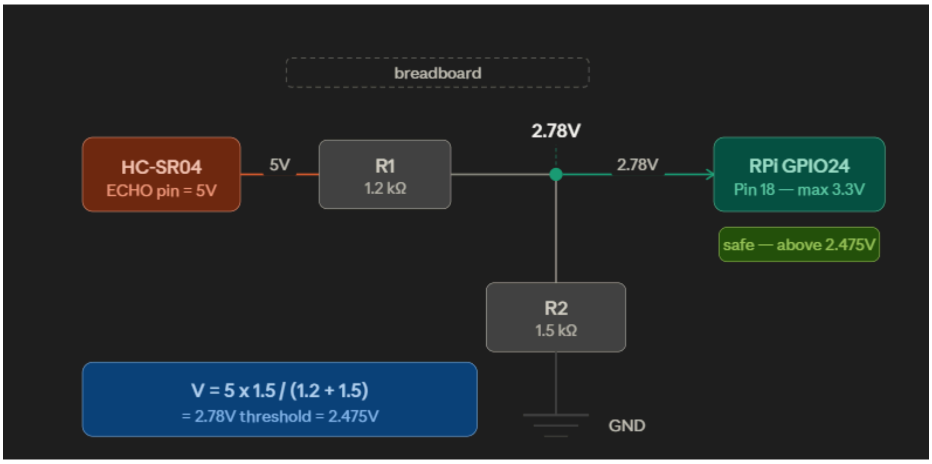
Component 4 — Raspberry Pi 4 (Model B, 8GB) alimentated with powerbank



Updated Full RPi GPIO Map (Component 4)

RPi Pin	GPIO	Connects to	Component	Notes
Pin 1	3.3V	VCC	Both wheel encoders	3.3V supply
Pin 6	GND	Common GND rail	—	Must share common ground with ESP32
Pin 8	GPIO14 TXD	GPIO3 / U0RXD	ESP32-CAM	UART TX → ESP32
Pin 10	GPIO15 RXD	GPIO1 / U0TXD	ESP32-CAM	UART RX ← ESP32
Pin 11	GPIO17	OUT	Left wheel encoder	✓ Interrupt capable — odometry left
Pin 12	**GPIO18**	**ENA**	**L298N**	✓ Hardware PWM — Left motor speed
Pin 16	GPIO23	TRIG	HC-SR04	3.3V output is enough ✓
Pin 18	**GPIO24**	**OUT**	**Right wheel encoder**	✓ Interrupt capable ← moved from Pin 12
Pin 22	**GPIO25**	**ECHO (via divider)**	**HC-SR04**	⚠ Never direct — divider mandatory ← moved from Pin 18
Pin 29	GPIO5	IN1	L298N	Left motor direction

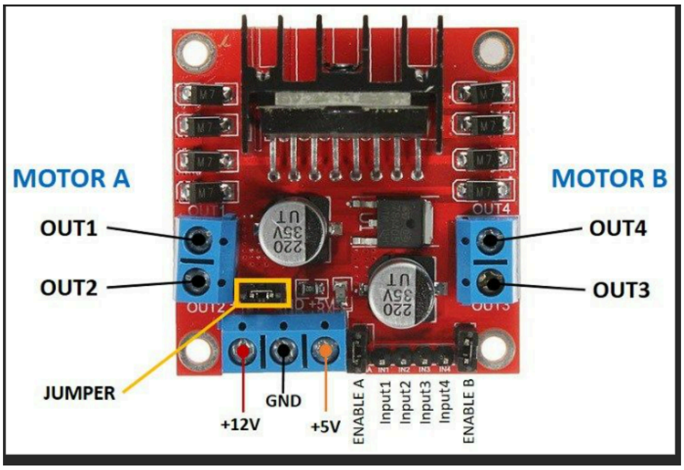
Pin 31	GPIO6	IN2	L298N	Left motor direction
Pin 32	GPIO12	Signal (orange wire)	MG996R servo	⚠ Hardware PWM — 1kΩ series resistor
Pin 33	GPIO13	IN3	L298N	Right motor direction ✅ unchanged
Pin 35	**GPIO19**	**ENB**	**L298N**	✓ Hardware PWM — Right motor speed
Pin 37	**GPIO26**	**IN4**	**L298N**	Right motor direction ← moved from Pin 35



*Voltage divider: R1 = 1.2kΩ between ECHO and **RPi GPIO25 (Pin 22)**. R2 = 1.5kΩ between **RPi GPIO25 (Pin 22)** and GND — Resulting voltage = 5 × 1.5/(1.2+1.5) = 2.78V*

Node	***Connects to***	***Notes***
R1 (1.2kΩ) pin 1	*HC-SR04 ECHO*	*Receives 5V from sensor*
R1 (1.2kΩ) pin 2	*RPi **GPIO25 (Pin 22)** AND R2 pin 1*	*Junction point — 2.78V here ✓*
R2 (1.5kΩ) pin 1	*R1 pin 2 AND RPi **GPIO25 (Pin 22)***	*Same junction*
R2 (1.5kΩ) pin 2	*Common GND rail*	*—*

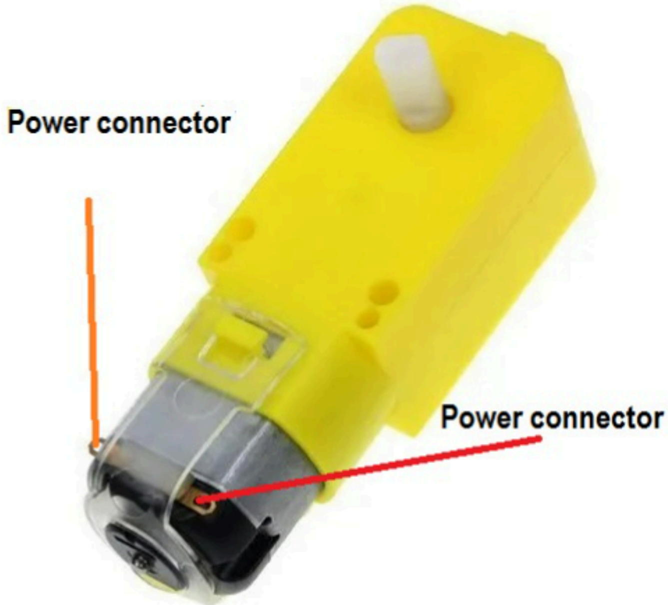
Component 5 — L298N dual H-bridge motor driver



Pin	Connects to	Component	Notes
VSS	(+) positive	LiPo 3S battery	Raw ~11.1V — direct, NOT regulated
VCC	VOUT	LM2596	5V logic supply
GND	Common GND rail	—	—
ENA	**GPIO18 (Pin 12)**	**Raspberry Pi 4**	✓ Hardware PWM — remove 5V bridge wire
ENB	**GPIO19 (Pin 35)**	**Raspberry Pi 4**	✓ Hardware PWM — remove 5V bridge wire
IN1	GPIO5 (Pin 29)	Raspberry Pi 4	Left motor direction
IN2	GPIO6 (Pin 31)	Raspberry Pi 4	Left motor direction
IN3	GPIO13 (Pin 33)	Raspberry Pi 4	Right motor direction
IN4	**GPIO26 (Pin 37)**	**Raspberry Pi 4**	Right motor direction ← moved
OUT1	Terminal A	Left DC motor	—

OUT2	Terminal B	Left DC motor	—
OUT3	Terminal A	Right DC motor	—
OUT4	Terminal B	Right DC motor	—

Component 6 — Left DC motor (yellow N20)



Pin	Connects to	Component	Notes
Terminal A	OUT1	L298N	—
Terminal B	OUT2	L298N	Swap A and B if motor spins in wrong direction

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Component 7 — Right DC motor (yellow N20)

Pin	Connects to	Component	Notes
Terminal A	OUT3	L298N	—
Terminal B	OUT4	L298N	Swap A and B if motor spins in wrong direction

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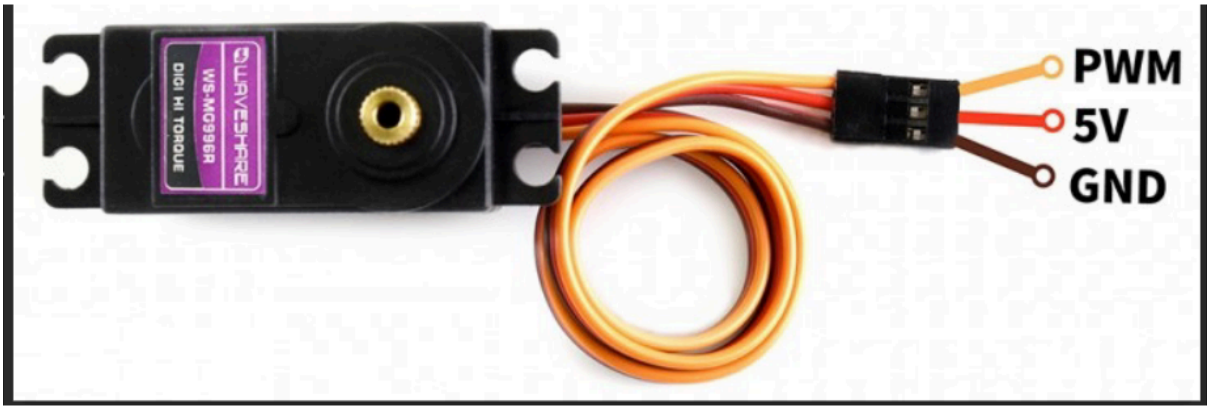
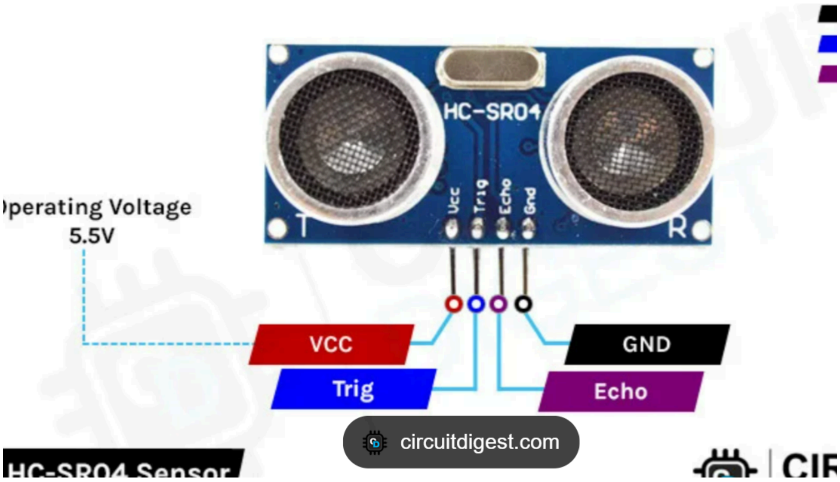
Component 8 — MG996R servo motor



Wire	Connects From	Through	Connects To	Notes
Signal (orange)	RPi Pin 32 (GPIO12)	1kΩ series resistor	MG996R Signal pin	⚠ Hardware PWM mandatory — no other RPi pin works for servo. 1kΩ resistor in series protects GPIO12
VCC (red)	LM2596 VOUT	—	MG996R VCC	5V supply — never from RPi 3.3V pin, draws too much current
GND (brown/black)	Common GND rail	—	MG996R GND	—

Component 9 — HC-SR04 ultrasonic sensor v2.0



The ECHO pin outputs 5V logic which will destroy the ESP32 GPIO (max 3.3V) if connected directly. A voltage divider is mandatory.

Voltage divider: R1 = 1kΩ between ECHO and ESP32-CAM GPIO18. R2 = 2kΩ between ESP32-CAM GPIO18 and GND Resulting voltage = 5 × 2/(1+2) = 3.33V ✓

Pin	Connects to	Component	Notes
VCC	VOUT	LM2596	5V supply
GND	Common GND rail	—	—
TRIG	GPIO23 (Pin 16)	Raspberry Pi 4	RPi drives TRIG directly — 3.3V output is enough ✓
ECHO	R1 (1.2kΩ) pin 1	Voltage divider	⚠ Never connect directly to RPi — 5V will damage GPIO , pi in output of divider

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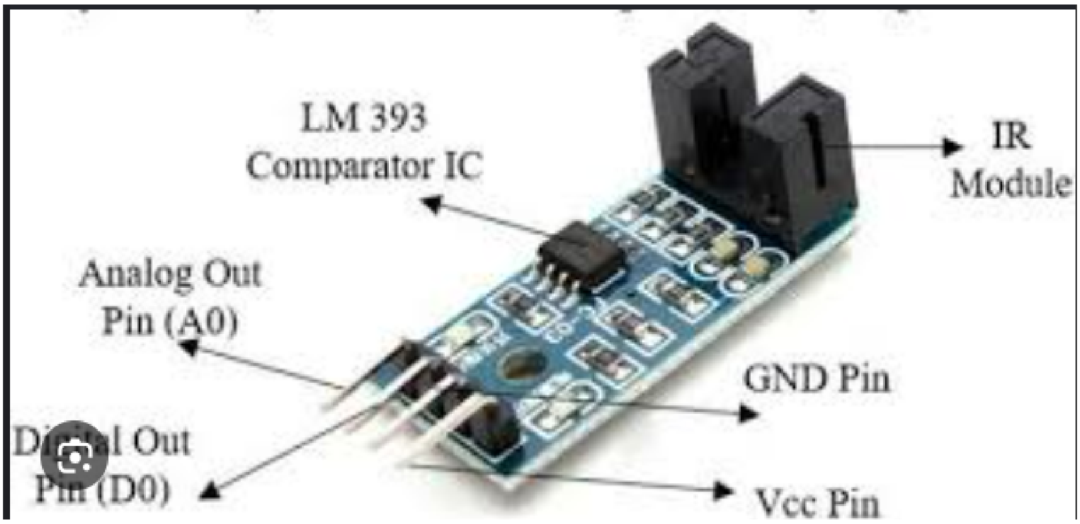
Component 10 — Left Wheel Encoder



Pin	Connects to	Component	Notes
VCC	Pin 1 (3.3V)	Raspberry Pi 4	3.3V supply — ensures signal level is RPi-safe
GND	Common GND rail	—	—
D0	GPIO17 (Pin 11)	Raspberry Pi 4	✓ Interrupt capable — left wheel odometry

A0 leave empty

Component 11 — Right Wheel Encoder

Pin	Connects to	Component	Notes
VCC	Pin 1 (3.3V)	Raspberry Pi 4	3.3V supply
GND	Common GND rail	—	—
D0	**GPIO24 (Pin 18)**	Raspberry Pi 4	✓ Interrupt capable ← moved from GPIO18